

Homework 1 - ET4369 Nyquist Rate Data Converters

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1 Anti-Alias Filtering

Given: Anti-alias filter $H(s)$ with 0 dB passband, -80 dB/decade roll-off beyond cutoff at $f_B = 1$ MHz, followed by an ideal sampler at rate f_s .

1.1 Part (a): Minimum f_s for > 70 dB alias rejection

After sampling at f_s , a signal component at frequency f_a aliases into the baseband. The worst-case in-band alias originates from frequency

$$f_a = f_s - f_B \quad (1)$$

which folds back to f_B (the edge of the signal band). Since this is the aliasing frequency closest to the passband, it experiences the least filter attenuation and therefore sets the requirement.

The filter attenuation at any frequency $f > f_B$ is

$$|H(f)|_{\text{dB}} = -80 \cdot \log_{10} \left(\frac{f}{f_B} \right) \quad (2)$$

Requiring > 70 dB of attenuation at $f_a = f_s - f_B$:

$$80 \cdot \log_{10} \left(\frac{f_s - f_B}{f_B} \right) > 70 \quad (3)$$

$$\log_{10} \left(\frac{f_s - f_B}{f_B} \right) > \frac{7}{8} \quad (4)$$

$$\frac{f_s - f_B}{f_B} > 10^{7/8} \quad (5)$$

$$f_s > f_B (1 + 10^{7/8}) \quad (6)$$

Evaluating numerically ($10^{7/8} \approx 7.499$):

$$\boxed{f_{s,\min} = (1 + 10^{7/8}) \times 1 \text{ MHz} \approx 8.50 \text{ MHz}} \quad (7)$$

The resulting oversampling ratio is

$$\boxed{\text{OSR} = \frac{f_s/2}{f_B} = \frac{8.50}{2 \times 1} \approx 4.25} \quad (8)$$

1.2 Part (b): Required OSR with -60 dB/decade roll-off

With a reduced roll-off of -60 dB/decade, the same analysis gives:

$$60 \cdot \log_{10} \left(\frac{f_s - f_B}{f_B} \right) > 70 \quad (9)$$

$$\log_{10} \left(\frac{f_s - f_B}{f_B} \right) > \frac{7}{6} \quad (10)$$

$$\frac{f_s - f_B}{f_B} > 10^{7/6} \quad (11)$$

$$f_s > f_B (1 + 10^{7/6}) \quad (12)$$

Evaluating numerically ($10^{7/6} = 10 \cdot 10^{1/6} \approx 14.68$):

$$f_{s,\min} = (1 + 14.68) \times 1 \text{ MHz} \approx 15.68 \text{ MHz} \quad (13)$$

$$\boxed{\text{OSR} = \frac{f_s/2}{f_B} = \frac{15.68}{2 \times 1} \approx 7.84} \quad (14)$$

2 State-of-the-Art Data Converters

Three recently published high-performance A/D converter papers from the *IEEE Journal of Solid-State Circuits* (2022) are reviewed below.

2.1 Paper 1: OTA-Free 1–1 MASH ADC

(a) IEEE Reference

S. T. Chandrasekaran, S. P. Bhanushali, S. Pietri and A. Sanyal, “OTA-Free 1–1 MASH ADC Using Fully Passive Noise-Shaping SAR & VCO ADC,” *IEEE J. Solid-State Circuits*, vol. 57, no. 4, pp. 1100–1111, Apr. 2022.

(b) Resolution, Sampling Rate, and Power

- SNDR: 71.5 dB $\Rightarrow$ ENOB $\approx$ 11.6 bits; Dynamic range: 75.8 dB
- Sampling rate: 24 MS/s (signal bandwidth 1.1 MHz at OSR = 11)
- Power consumption: 0.16 mW from a 1.1 V supply

(c) Technology and Chip Area

- Technology: 65 nm CMOS
- Active core area: 0.045 mm²

(d) Architecture and Application

- Architecture: 1–1 MASH oversampling ADC with a fully passive noise-shaping SAR as the first stage and a ring VCO as the second stage. The design is OTA-free and highly digital/scaling-friendly.
- Application: Low-power, medium-bandwidth applications such as sensor interfaces and IoT devices.

(e) Battery Life Estimate

$$t = \frac{E_{\text{battery}}}{P} = \frac{5 \text{ Wh}}{0.16 \text{ mW}} = 31,250 \text{ hours} \approx \mathbf{1,302} \text{ days} \approx \mathbf{3.6} \text{ years}$$

2.2 Paper 2: 7-Bit Two-Step Flash ADC

(a) IEEE Reference

D.-R. Oh, M.-J. Seo and S.-T. Ryu, “A 7-Bit Two-Step Flash ADC With Sample-and-Hold Sharing Technique,” *IEEE J. Solid-State Circuits*, vol. 57, no. 9, pp. 2791–2801, Sep. 2022.

(b) Resolution, Sampling Rate, and Power

- Resolution: 7 bits (ENOB 6.34 at Nyquist input)
- Sampling rate: 3 GS/s (two-channel time-interleaved)
- Power consumption: 6.8 mW from a 0.9 V supply

(c) Technology and Chip Area

- Technology: 40 nm CMOS
- Core area: 0.03 mm² (including offset calibration circuitry)

(d) Architecture and Application

- Architecture: Two-step flash ADC (2.5-bit coarse + 5-bit fine) with sample-and-hold sharing technique and reference-embedding eight-time interpolating fine ADC. Two-channel time-interleaved for 3 GS/s operation.
- Application: High-speed wireline/wireless communication receivers requiring wide effective resolution bandwidth (ERBW up to 7 GHz).

(e) Battery Life Estimate

$$t = \frac{5 \text{ Wh}}{6.8 \text{ mW}} = 735.3 \text{ hours} \approx \mathbf{30.6} \text{ days}$$

2.3 Paper 3: SAR-Assisted Noise-Shaping Pipeline ADC

(a) IEEE Reference

Y. Zhang, J. Zhang, S. Liu, R. Ding, Y. Zhu, C.-H. Chan and R. P. Martins, “A 20 MHz Bandwidth 79 dB SNDR SAR-Assisted Noise-Shaping Pipeline ADC With Gain and Offset Calibrations,” *IEEE J. Solid-State Circuits*, vol. 57, no. 3, pp. 745–756, Mar. 2022.

(b) Resolution, Sampling Rate, and Power

- SNDR: 79.1 dB $\Rightarrow$ ENOB $\approx$ 12.8 bits; Bandwidth: 20 MHz
- Sampling rate: 260 MS/s (OSR = 6.5)
- Power consumption: 3.1 mW from a 1 V supply

(c) Technology and Chip Area

- Technology: 28 nm CMOS
- Active area: 0.037 mm²

(d) Architecture and Application

- Architecture: Second-order SAR-assisted noise-shaping pipeline ADC with error-feedback (EF) structure, optimized NTF zeros, background inter-stage gain and offset calibrations, and partial interleaving.
- Application: Wideband communication baseband processing and software-defined radio.

(e) Battery Life Estimate

$$t = \frac{5 \text{ Wh}}{3.1 \text{ mW}} = 1,612.9 \text{ hours} \approx \mathbf{67.2} \text{ days}$$

2.4 Summary Comparison

Table 1: Comparison of the three selected ADCs.

Parameter	Paper 1	Paper 2	Paper 3
Architecture	NS-SAR + VCO MASH	Two-step flash (TI)	NS pipeline (SAR-assisted)
Technology	65 nm	40 nm	28 nm
Resolution	11.6-bit ENOB	7-bit (6.34 ENOB)	12.8-bit ENOB
Sampling rate	24 MS/s	3 GS/s	260 MS/s
Bandwidth	1.1 MHz	1.5 GHz (Nyquist)	20 MHz
Power	0.16 mW	6.8 mW	3.1 mW
Area	0.045 mm ²	0.03 mm ²	0.037 mm ²
Supply	1.1 V	0.9 V	1 V
Battery life	≈3.6 yr	≈30.6 days	≈67.2 days

3 Quantization Noise Power Calculation for a Ramp Input Signal

3.1 Part (a): Ideal 3-bit Mid-Tread Quantizer

3.1.1 Setup

An ideal 3-bit mid-tread quantizer has $2^3 = 8$ output levels at $D_{\text{out}} \in \{-4, -3, -2, -1, 0, 1, 2, 3\}$ (in units of Δ), where Δ is the quantization step size. The decision boundaries are at $\{-3.5, -2.5, -1.5, -0.5, 0.5, 1.5, 2.5\}\Delta$, and the full-scale range is $\text{FSR} = 2^B \cdot \Delta = 8\Delta$, spanning the input range $[-4.5\Delta, 3.5\Delta]$.

The ramp input is:

$$x(t) = -4.5\Delta + 8\Delta \cdot \frac{t}{t_1}, \quad t \in [0, t_1] \quad (15)$$

3.1.2 Quantization Error

Within each quantization step centered at output level $k\Delta$, the input x ranges over an interval of width Δ . The quantization error is:

$$e(x) = Q(x) - x = k\Delta - x \quad (16)$$

which is a linear function that decreases from $+\Delta/2$ to $-\Delta/2$ as x traverses the step. Since the ramp input sweeps the full scale range uniformly, the error signal is a **periodic sawtooth wave** with 8 complete periods (one per quantization step), each with peak-to-peak amplitude Δ .

The quantization error as a function of normalized time is shown in Figure 1.

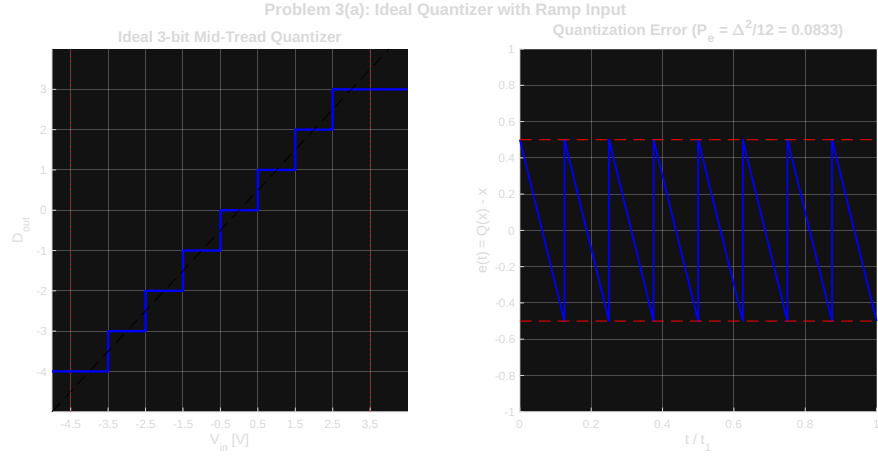


Figure 1: Ideal 3-bit mid-tread quantizer: transfer characteristic (left) and quantization error for ramp input (right).

3.1.3 Noise Power Calculation

The quantization noise power is the mean-square value of the error over the full ramp:

$$P_e = \frac{1}{\text{FSR}} \int_{-4.5\Delta}^{3.5\Delta} e^2(x) dx \quad (17)$$

Since each of the 8 steps contributes an identical integral, we compute the integral for one step. For a step centered at $k\Delta$ with input range $[(k - \frac{1}{2})\Delta, (k + \frac{1}{2})\Delta]$:

$$\int_{(k - \frac{1}{2})\Delta}^{(k + \frac{1}{2})\Delta} (k\Delta - x)^2 dx = \int_{-\Delta/2}^{\Delta/2} u^2 du = \left[\frac{u^3}{3} \right]_{-\Delta/2}^{\Delta/2} = \frac{2}{3} \left(\frac{\Delta}{2} \right)^3 = \frac{\Delta^3}{12} \quad (18)$$

Summing over all 8 steps:

$$P_e = \frac{1}{8\Delta} \cdot 8 \cdot \frac{\Delta^3}{12} = \boxed{\frac{\Delta^2}{12}} \quad (19)$$

This is **exactly** the standard quantization noise model result $\Delta^2/12$, confirmed numerically by MATLAB ($P_e = 0.0833$).

3.1.4 Discussion

The ramp input is special: it traverses each quantization bin uniformly, spending equal time in every step. The resulting error is a perfect sawtooth wave, equivalent to a uniform distribution on $[-\Delta/2, +\Delta/2]$, which is exactly the assumption of the statistical quantization noise model.

Why didn't we use this approach for the sinusoidal case? A sinusoidal input does *not* traverse quantization bins uniformly. Near its peaks ($\pm A$), the sinusoid's slope is small, so it spends more time in those bins; near zero crossings, it moves quickly. The input probability density of a sinusoid is the arc-sine distribution, not uniform. Therefore, the deterministic sawtooth analysis does not directly apply to a sinusoid, and one must instead rely on the statistical approximation (assuming the quantization error is uniformly distributed and uncorrelated with the signal), which holds well when the signal spans many quantization levels and the frequency is not harmonically related to the sampling frequency.

3.2 Part (b): Broken Quantizer (Missing Codes 0 and 1)

3.2.1 Identifying the Defect

From the given figure, the “broken” quantizer has the following transfer characteristic:

Output Code D_{out}	Input Range V_{in}	Step Width
-4	$[-4.5\Delta, -3.5\Delta)$	Δ
-3	$[-3.5\Delta, -2.5\Delta)$	Δ
-2	$[-2.5\Delta, -1.5\Delta)$	Δ
-1	$[-1.5\Delta, +1.5\Delta)$	3Δ
+2	$[+1.5\Delta, +2.5\Delta)$	Δ
+3	$[+2.5\Delta, +3.5\Delta)$	Δ

Table 2: Transfer characteristic of the broken quantizer. Codes 0 and 1 are missing.

The step at $D_{\text{out}} = -1$ is **three times wider** than normal (3Δ instead of Δ), absorbing the input ranges that would normally correspond to codes 0 and 1.

3.2.2 Quantization Error

The quantization error and its sketch are shown in Figure 2.

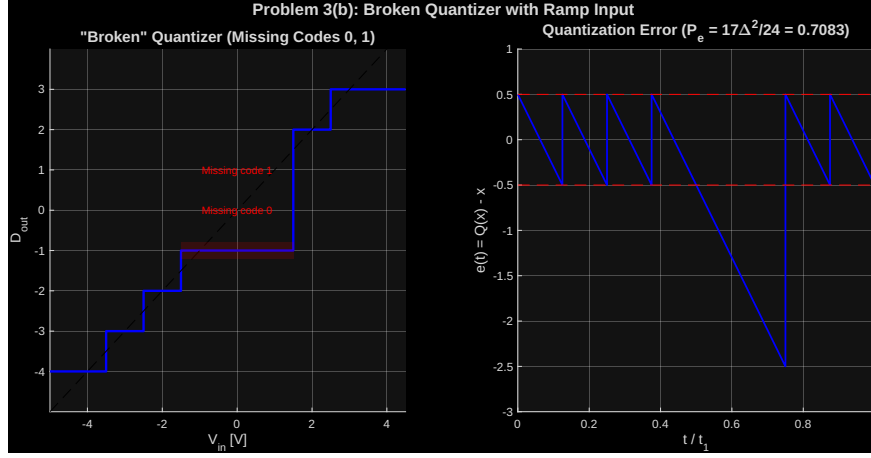


Figure 2: Broken quantizer: transfer characteristic (left) and quantization error for ramp input (right). The error greatly exceeds $\pm\Delta/2$ during the wide step.

For the 5 normal-width steps, the error is the standard sawtooth from $+\Delta/2$ to $-\Delta/2$. For the triple-width step at $D_{\text{out}} = -1$:

$$e(x) = (-1)\Delta - x, \quad x \in [-1.5\Delta, 1.5\Delta) \quad (20)$$

- At $x = -1.5\Delta$: $e = -\Delta - (-1.5\Delta) = +\frac{\Delta}{2}$
- At $x = +1.5\Delta$: $e = -\Delta - 1.5\Delta = -\frac{5\Delta}{2}$

The error grows far beyond the normal $\pm\Delta/2$ bounds, reaching $-5\Delta/2$ at the end of the wide step.

3.2.3 Noise Power Calculation

The total noise power is:

$$P_e = \frac{1}{\text{FSR}} \left[\sum_{5 \text{ normal steps}} \int e^2 dx + \int_{\text{wide step}} e^2 dx \right] \quad (21)$$

Normal steps (5 steps of width Δ): each contributes $\Delta^3/12$:

$$\sum_{\text{normal}} \int e^2 dx = 5 \cdot \frac{\Delta^3}{12} = \frac{5\Delta^3}{12} \quad (22)$$

Wide step (output = $-\Delta$, input $\in [-1.5\Delta, 1.5\Delta]$):

$$\begin{aligned}
\int_{-1.5\Delta}^{1.5\Delta} (-\Delta - x)^2 dx &= \int_{-1.5\Delta}^{1.5\Delta} (\Delta^2 + 2\Delta x + x^2) dx \\
&= \left[\Delta^2 x + \Delta x^2 + \frac{x^3}{3} \right]_{-1.5\Delta}^{1.5\Delta} \\
&= \left(1.5\Delta^3 + 2.25\Delta^3 + \frac{3.375\Delta^3}{3} \right) - \left(-1.5\Delta^3 + 2.25\Delta^3 - \frac{3.375\Delta^3}{3} \right) \\
&= (1.5 + 2.25 + 1.125)\Delta^3 - (-1.5 + 2.25 - 1.125)\Delta^3 \\
&= 4.875\Delta^3 - (-0.375)\Delta^3 = \frac{21\Delta^3}{4} \tag{23}
\end{aligned}$$

Total noise power:

$$P_e = \frac{1}{8\Delta} \left(\frac{5\Delta^3}{12} + \frac{21\Delta^3}{4} \right) = \frac{1}{8\Delta} \cdot \frac{5\Delta^3 + 63\Delta^3}{12} = \frac{68\Delta^2}{96} = \boxed{\frac{17\Delta^2}{24}} \tag{24}$$

This result is confirmed numerically by MATLAB ($P_e = 0.7083$).

3.2.4 Comparison

The broken quantizer has a noise power of $17\Delta^2/24$, compared to $\Delta^2/12$ for the ideal quantizer:

$$\frac{P_{e,\text{broken}}}{P_{e,\text{ideal}}} = \frac{17\Delta^2/24}{\Delta^2/12} = \frac{17}{2} = 8.5 \tag{25}$$

4 Quantization Noise Model Validation

4.1 Signal Generation

Four test signals of 10,000 samples are generated, all with equal power $P = A^2/2 = 0.5$:

1. **Sinusoidal:** $x_1(n) = A \cos(2\pi \cdot 67/2048 \cdot n)$, $A = 1$
2. **Uniform:** $x_2 \sim \mathcal{U}[-a, a]$, $a = \sqrt{3P} = \sqrt{3/2} \approx 1.225$
3. **Gaussian:** $x_3 \sim \mathcal{N}(0, \sigma^2)$, $\sigma = \sqrt{P} = 1/\sqrt{2} \approx 0.707$
4. **ReLU Gaussian:** $x_4 = \max(\mathcal{N}(0, \sigma_{hw}^2), 0)$, $\sigma_{hw} = \sqrt{2P} = 1$

The power-matching conditions follow from:

$$\text{Uniform: } E[x^2] = \frac{a^2}{3} = P \Rightarrow a = \sqrt{3P} \tag{26}$$

$$\text{Gaussian: } E[x^2] = \sigma^2 = P \Rightarrow \sigma = \sqrt{P} \tag{27}$$

$$\text{ReLU Gaussian: } E[x^2] = \frac{\sigma_{hw}^2}{2} = P \Rightarrow \sigma_{hw} = \sqrt{2P} \tag{28}$$

4.2 SQNR vs Loading Factor ($B = 10$)

Figure 3 shows the range-precision trade-off for all four signals at $B = 10$ bits. The loading factor is defined as $LF = (V_{FSR}/2)/x_{rms}$.

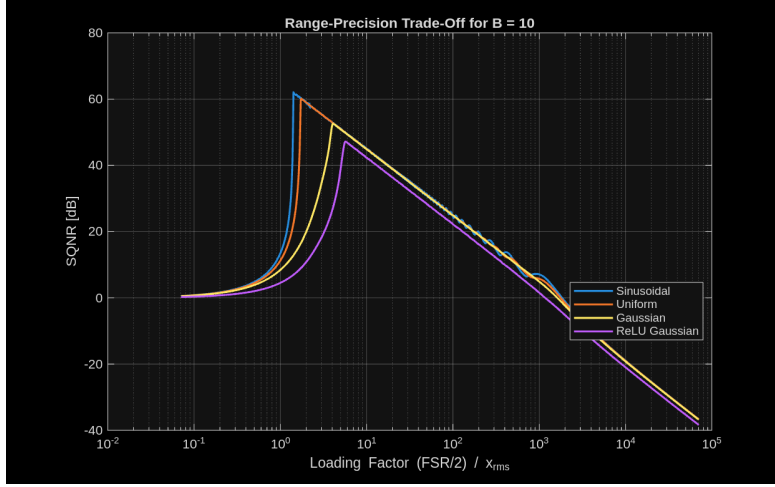


Figure 3: SQNR vs loading factor for all four input signals at $B = 10$.

At low loading factors the quantizer overloads (clipping), degrading SQNR. At high loading factors the signal occupies only a few quantization levels, wasting resolution. The sinusoidal input achieves the highest peak SQNR because its bounded amplitude avoids clipping at the optimal loading factor. The Gaussian and ReLU Gaussian signals, having unbounded tails, require a larger FSR to avoid clipping and therefore achieve lower peak SQNR.

4.3 SQNR vs Loading Factor Across Resolutions (Sinusoidal)

Figure 4 shows how the SQNR curve shifts upward by approximately 6 dB per additional bit, consistent with the theoretical $SQNR = 6.02B + 1.76$ dB for a full-scale sinusoid.

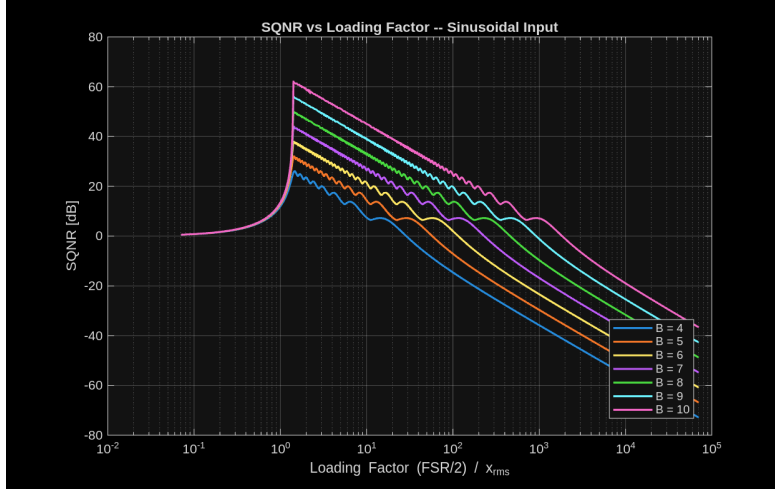


Figure 4: SQNR vs loading factor for sinusoidal input at resolutions $B = 4 \dots 10$.

4.4 Optimal SQNR and Loading Factor vs Resolution

Figure 5 shows the peak SQNR and the corresponding optimal loading factor as a function of bit resolution for each signal type.

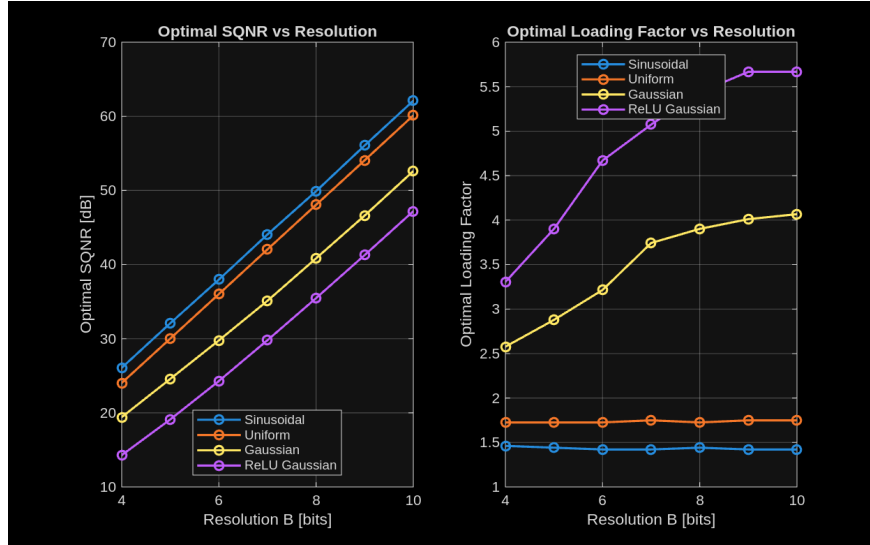


Figure 5: Left: optimal SQNR vs resolution. Right: optimal loading factor vs resolution.

4.5 Observations

1. The sinusoidal input achieves the highest SQNR at each resolution because its bounded amplitude allows the quantizer to be loaded optimally without clipping.
2. The Gaussian input requires a larger loading factor (wider FSR) to accommodate its tails, trading quantization precision for reduced clipping distortion.
3. The ReLU Gaussian, being unipolar, is less efficiently matched to the bipolar mid-rise quantizer, resulting in the lowest SQNR.
4. For all signals, SQNR increases by ≈ 6 dB per bit at the optimal loading factor, confirming the $6.02B$ rule of thumb.
5. The optimal loading factor is signal-dependent and approximately constant across resolutions for each type of signal.

5 SQNR Calculation

5.1 Setup

- Ideal quantizer, $B = 14$ bits
- Input: periodic full-scale half-wave rectified sinusoid with amplitude A and period T
- Statistical quantizer model applies (sufficiently randomized samples)

5.2 Signal Power

The half-wave rectified sinusoid is defined as:

$$v_{in}(t) = \begin{cases} A \sin\left(\frac{2\pi t}{T}\right) & 0 \leq t \leq T/2 \\ 0 & T/2 < t \leq T \end{cases}$$

Its time-averaged power is:

$$\begin{aligned} P_{sig} &= \frac{1}{T} \int_0^T v_{in}^2(t) dt = \frac{1}{T} \int_0^{T/2} A^2 \sin^2\left(\frac{2\pi t}{T}\right) dt \\ &= \frac{A^2}{2T} \int_0^{T/2} \left[1 - \cos\left(\frac{4\pi t}{T}\right)\right] dt = \frac{A^2}{2T} \left[\frac{T}{2} - 0\right] \end{aligned} \quad (29)$$

$$\boxed{P_{sig} = \frac{A^2}{4}}$$

5.3 Quantization Noise Power

The signal is unipolar, spanning $[0, A]$. Since it is *full-scale*, the quantizer's full-scale range matches the signal range:

$$V_{FS} = A, \quad \Delta = \frac{V_{FS}}{2^B} = \frac{A}{2^{14}}$$

Under the statistical model, the quantization noise power is:

$$P_q = \frac{\Delta^2}{12} = \frac{A^2}{12 \cdot 2^{28}}$$

5.4 SQNR Calculation

$$\text{SQNR} = \frac{P_{sig}}{P_q} = \frac{A^2/4}{A^2/(12 \cdot 2^{28})} = \frac{12 \cdot 2^{28}}{4} = 3 \cdot 2^{28} \quad (30)$$

Converting to dB:

$$\begin{aligned} \text{SQNR}_{\text{dB}} &= 10 \log_{10}(3 \cdot 2^{28}) \\ &= 10 \log_{10}(3) + 28 \times 10 \log_{10}(2) \\ &= 4.77 + 84.29 \end{aligned} \quad (31)$$

$\text{SQNR} \approx 89.1 \text{ dB}$

5.5 Comparison with Standard Full-Scale Sinusoid

For a full-scale sinusoid on a bipolar quantizer, $\text{SQNR} = 6.02B + 1.76 = 86.0 \text{ dB}$ at $B = 14$.

The half-wave rectified signal achieves 3 dB *higher* SQNR despite having half the power of a full sinusoid ($A^2/4$ vs. $A^2/2$). This is because the unipolar signal only requires half the quantizer range ($V_{FS} = A$ instead of $2A$), reducing Δ by a factor of 2 and thus P_q by a factor of 4 (+6 dB). The net effect is $-3 + 6 = +3 \text{ dB}$ relative to the bipolar sinusoidal case.